

As explained, a prerequisite for establishing SSL communication is having a web server with an SSL certificate (SSL server). Here are the key processes of the SSL authentication mechanism:

1. A client sends a request to the SSL server
2. When the SSL server receives the request, the SSL server sends its SSL certificate with its *public key* to the client
3. Once the client receives the SSL certificate and verifies it, the client creates a symmetric key and encrypts the *symmetric key* using the *public key* from the server
4. Then, the client sends the encrypted *symmetric key* to the SSL server
5. When the SSL server receives the encrypted *symmetric key*, the SSL server decrypts it with its own *private key*. By now, the client and the SSL server have the same *symmetric key* sent through a secured channel.
6. Both use the *symmetric key* as a session key to create a secured communication channel

As you can see in the processes above, SSL uses both **Asymmetric Encryption** and **Symmetric Encryption**. SSL uses Asymmetric Encryption for the first authentication, which is more secure. For a session key, SSL uses Symmetric Encryption, which is faster.

How to set up SSL using Certbot

Certbot allows you to get an SSL certificate on your web server easily and quickly.

Here are the key steps:

1. Install Certbot for Nginx on Ubuntu OS
2. Open an SSL port
3. Obtain an SSL certificate by running the certbot command